FACIALPULSE

CLINICAL MULTIMODAL AFFECT ANALYSIS & STIMULUS-RESPONSE DEPRESSION SCREENING PLATFORM

Image and Video Processing Project

Presented by: Group 9

Technology: Next.js, FastAPI, MediaPipe, TensorFlow Lite and Machine Learning

Project Team

Group 9

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Literature Survey

Existing Approaches and Research Direction

Facial Action Unit analysis	Studies facial muscle activity and expression changes.
Acoustic speech analysis	Studies pitch, pauses, voice variation and speech patterns.
Multimodal depression assessment	Combines visual and audio features for symptom estimation.

Limitations addressed by the project:

- Static facial images do not capture changes in expression over time.
- Natural facial structure can differ between individuals.
- Facial expressions alone may not represent the complete affective response.
- Audio and visual features can provide complementary information.

Research Papers

1. “An Automated and Real-time Approach of Depression Detection from Facial Micro-expressions” — Ghulam Gilanie et al. (2022).

2. “Automated Facial Video-Based Recognition of Depression and Anxiety Symptom Severity: Cross-Corpus Validation” — A. Pampouchidou et al. (2020).

3. “FacePsy: An Open-Source Affective Mobile Sensing System – Analyzing Facial Behavior and Head Gesture for Depression Detection in Naturalistic Settings” — Rahul Islam and Sang Won Bae (2024).

Gilanie et al. (2022) – Automated Real-time Depression Detection from Facial Micro-expressions

Real-time detection using facial micro-expressions and FACS-based muscle movement analysis

CNN and SVM classification
CNN and SVM classification
PHQ-8 for clinical assessment

Dataset

180 clinically depressed patients
200 healthy controls
24 FPS videos, 3–5 minutes each

Results

CNN accuracy

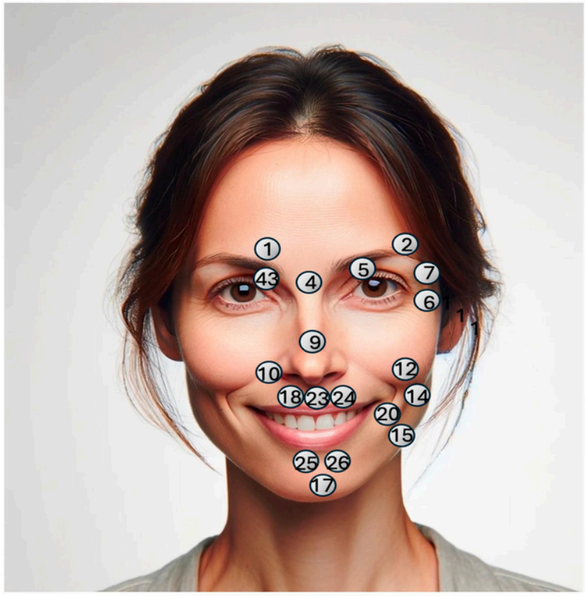
99.9%

SVM accuracy

100%

Limitations

Hospital-based dataset
Limited real-world generalizability
Facial data only, no multimodal inputs



AU01 Inner Brow Raiser
AU02 Outer Brow Raiser
AU04 Brow Lowerer
AU05 Upper Lid Raiser
AU06 Cheek Raiser
AU07 Lid Tightener
AU09 Nose Wrinkler
AU10 Upper Lid Raiser
AU12 Lip Corner Pull
AU14 Dimpler
AU15 Lip Corner Depressor
AU17 Chin Raiser
AU18 Lip Puckerer
AU20 Lip Stretcher
AU23 Lip Tightener
AU24 Lip Pressor
AU25 Lips Part
AU26 Jaw Drop
AU43 Eyes Closed

Pampouchidou et al. (2020) – Automated Facial Video-Based Recognition of Depression and Anxiety Severity

Objective: estimate depression and anxiety symptom severity via facial video analysis and evaluate cross-dataset generalization

APPROACH

MOTION FEATURES

Motion History Images (MHI)

→

APPEARANCE FEATURES

Local Binary Patterns (LBP) and HOG

→

DEEP FEATURES

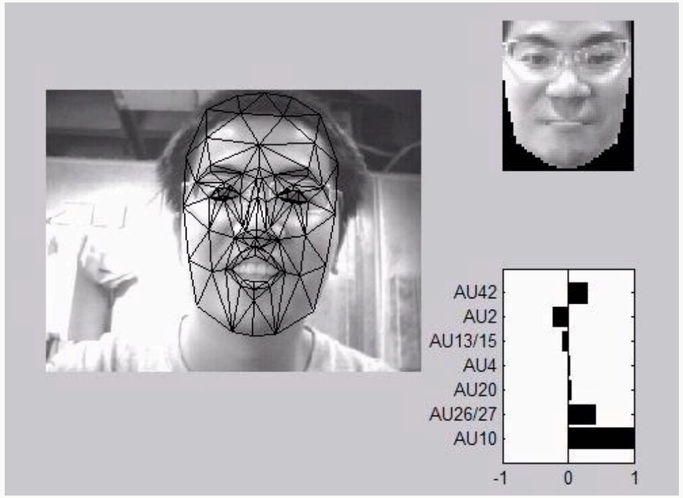
VGG transfer learning

→

MODEL & PREDICTION

Dimensionality reduction and machine learning

Binary classification and continuous severity prediction



DATASETS

20 clinically diagnosed depressed patients and 45 healthy volunteers

Additional cross-corpus evaluation on AVEC'14 dataset

KEY FINDINGS

Continuous severity estimation outperformed binary classification

Comparable performance on AVEC'14 dataset

Visual indicators more accurate for self reported anxiety than expert-rated depression

LIMITATIONS

Small clinical dataset

Overlap between depression and anxiety facial indicators

Need for larger, more diverse populations for validation

Islam and Bae (2024) – FacePsy

Open-Source Mobile Sensing for Depression Detection in Naturalistic Settings

OBJECTIVE

Develop open-source mobile sensing system to monitor facial behavior and head gestures in everyday naturalistic environments

Detect depressive episodes outside laboratory settings

METHODOLOGY

Facial capture via smartphone camera

Extract facial landmarks, Action Units, eye-open states, smiles, and head gestures

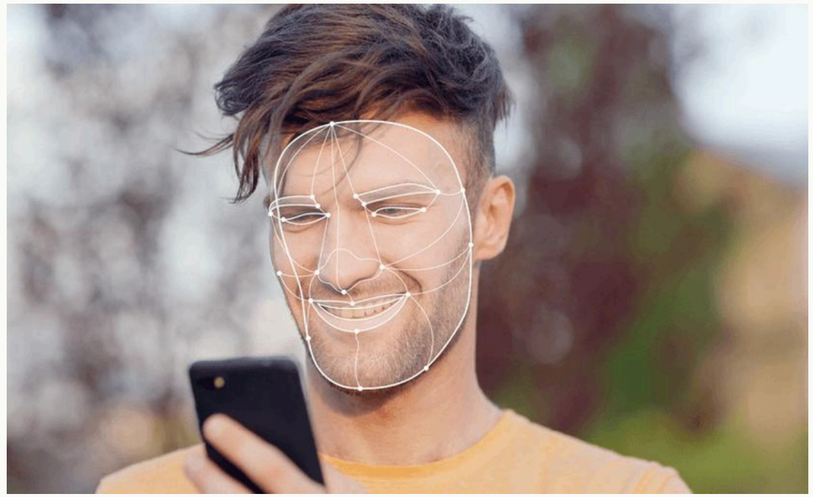
Local feature extraction on device, discarding raw images for privacy

Machine-learning models predict depressive episodes

DATASET

25 participants over four weeks

544 days of facial data collected via smartphone triggers



KEY FINDINGS

Significant AUs identified: AU2, AU6, AU7, AU12, AU15, AU17

Hybrid model AUROC reported as 81%

PHQ-9 score prediction MAE of 3.08

LIMITATIONS

Small participant size

Variable data availability across days

Need for more longitudinal data to improve performance

Comparative Analysis and Research Gap – Motivating FacialPulse

Study	Pampouchidou et al. (2020)	Jain et al. (2021)	Alghowinem et al. (2022)
PRIMARY FOCUS	Depression and anxiety severity estimation	Depression level classification	Depression detection using facial analysis
FACIAL FEATURES USED	Motion history images (MHI), LBP, HOG	Facial Action Units (AUs)	Facial landmarks and geometric features
MACHINE LEARNING APPROACH	VGG transfer learning, binary classification	CNN with VGG-Face feature extraction	CNN with custom feature learning
ENVIRONMENT	Clinical, controlled setting	Controlled lab environment	Controlled database conditions
MAIN CONTRIBUTION	Cross-dataset evaluation of facial features for severity estimation	AU-based depression classification with deep learning	Depression detection using facial landmark geometry

RESEARCH GAPS IDENTIFIED

- Heavy reliance on visual facial features alone
- Facial indicators may not fully capture true emotional state
- Real-world performance and generalizability remain challenging
- Continuous temporal tracking insufficiently explored
- Privacy concerns and lack of multimodal integration

HOW FACIALPULSE ADDRESSES THESE GAPS

- Multimodal fusion combining facial Action Units and acoustic speech features
- Temporal smoothing via 30-frame sliding window and exponential moving average
- Structured 43-second stimulus response protocol for consistent assessment
- Machine-learning PHQ-8 symptom severity scoring
- Longitudinal tracking across multiple sessions for continuous monitoring

Problem Statement

Traditional approaches to depression assessment may depend heavily on questionnaires or subjective observation. Static facial analysis also fails to capture how expressions change in response to emotional and cognitive stimuli.

Objectives

1. Analyze facial expressions using Action Units.
2. Establish an individual resting facial baseline.
3. Measure changes in facial activity over time.
4. Extract acoustic features from speech.
5. Combine facial and audio features.
6. Estimate PHQ-8 symptom scores using an ML-based scoring pipeline.
7. Support longitudinal tracking of assessment sessions.
8. Protect user privacy through pseudonymization and raw-media handling.

What is FacialPulse?

FacialPulse is an AI-powered clinical screening platform designed to analyze facial expressions and speech characteristics during a guided assessment.
It combines three major components:

- **Facial analysis**: Detects facial Action Units and their temporal changes.
- **Acoustic analysis**: Extracts speech prosody and voice-related features.
- **Machine learning**: Combines extracted features to estimate PHQ-8 depression symptom scores.

Main objective: To study affective responses and speech patterns through an **interactive stimulus-response assessment** rather than relying only on static facial images.
The platform is intended for adjunctive screening and research, not as a standalone diagnostic system.

Technical Foundations

- MediaPipe Face Landmarker
- Next.js 14
- TensorFlow Lite
- Web Audio API
- OpenCV
- Machine Learning
- FastAPI

Proposed Methodology

Phase 1 — Temporal Analysis and Baseline Calibration

Facial feature smoothing and individual baseline establishment

↓

Phase 2 — Guided Elicitation Protocol

Four-stage stimulus-response assessment

↓

Phase 3 — Audio-Visual Feature Analysis

Acoustic biomarkers and multimodal fusion

↓

Phase 4 — Machine Learning Scoring

PHQ-8 score estimation and feature attribution

↓

Phase 5 — Privacy and Longitudinal Tracking

Pseudonymized history, reports and data deletion

Facial Action Unit Analysis

AU01	Inner brow raiser
AU04	Brow lowerer
AU06	Cheek raiser
AU12	Lip corner puller
AU15	Lip corner depressor
AU45	Blink

Image preprocessing

1 Grayscale conversion
2 Median blur
3 Histogram equalization
4 Resize to 200 × 200
5 Normalize pixel values

TensorFlow Lite classifier · AU_200.tflite

Temporal Analysis and Guided Protocol

Temporal Analysis

- 30-frame sliding window
- Exponential Moving Average smoothing
- EMA coefficient: α = 0.35
- Individual baseline calibration

Guided 43-Second Assessment

Resting baseline	8 seconds	Measures neutral facial state
Positive affect	12 seconds	Measures smile intensity and onset
Rainbow Passage reading	15 seconds	Studies speech and facial activity under cognitive load
Affective recovery	8 seconds	Measures recovery and residual tension

Baseline target: 8 frames;
quick neutral calibration: 4 seconds

Total duration: 43 seconds

Acoustic Feature Extraction

16 kHz

audio sample rate

Fundamental Frequency

Pitch standard deviation
Semitone dispersion
Monotone index

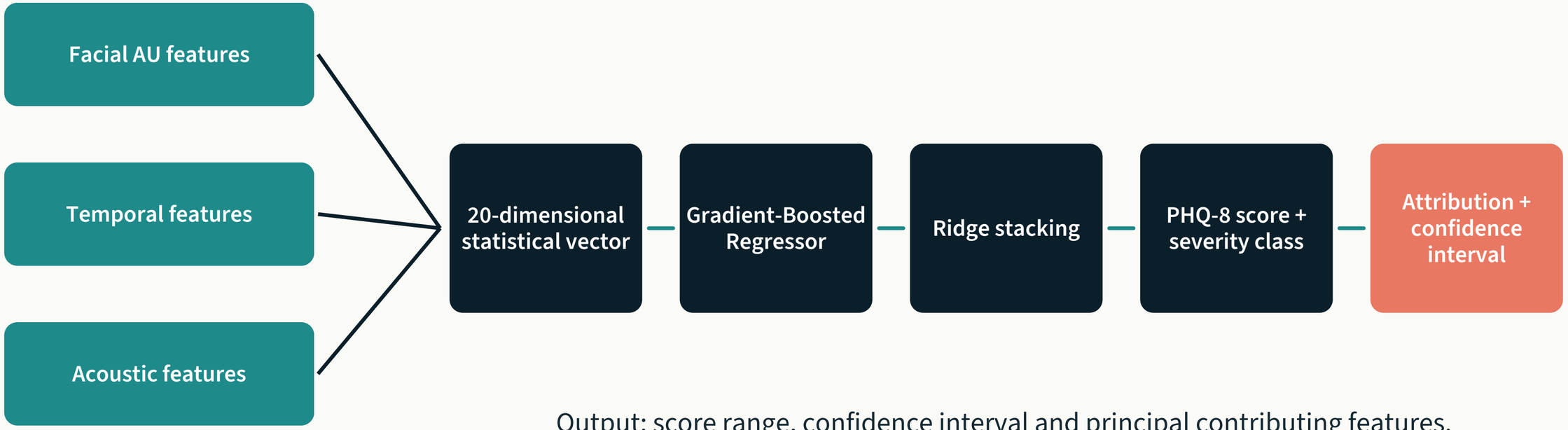
Speech Activity

Voice Activity Detection
Speech-to-pause ratio
Pause latency > 750 ms

Voice Quality

Jitter
Shimmer
Vocal strain index

Multimodal Fusion and ML Scoring



Output: score range, confidence interval and principal contributing features.

Dataset and Benchmark References

DAIC-WOZ

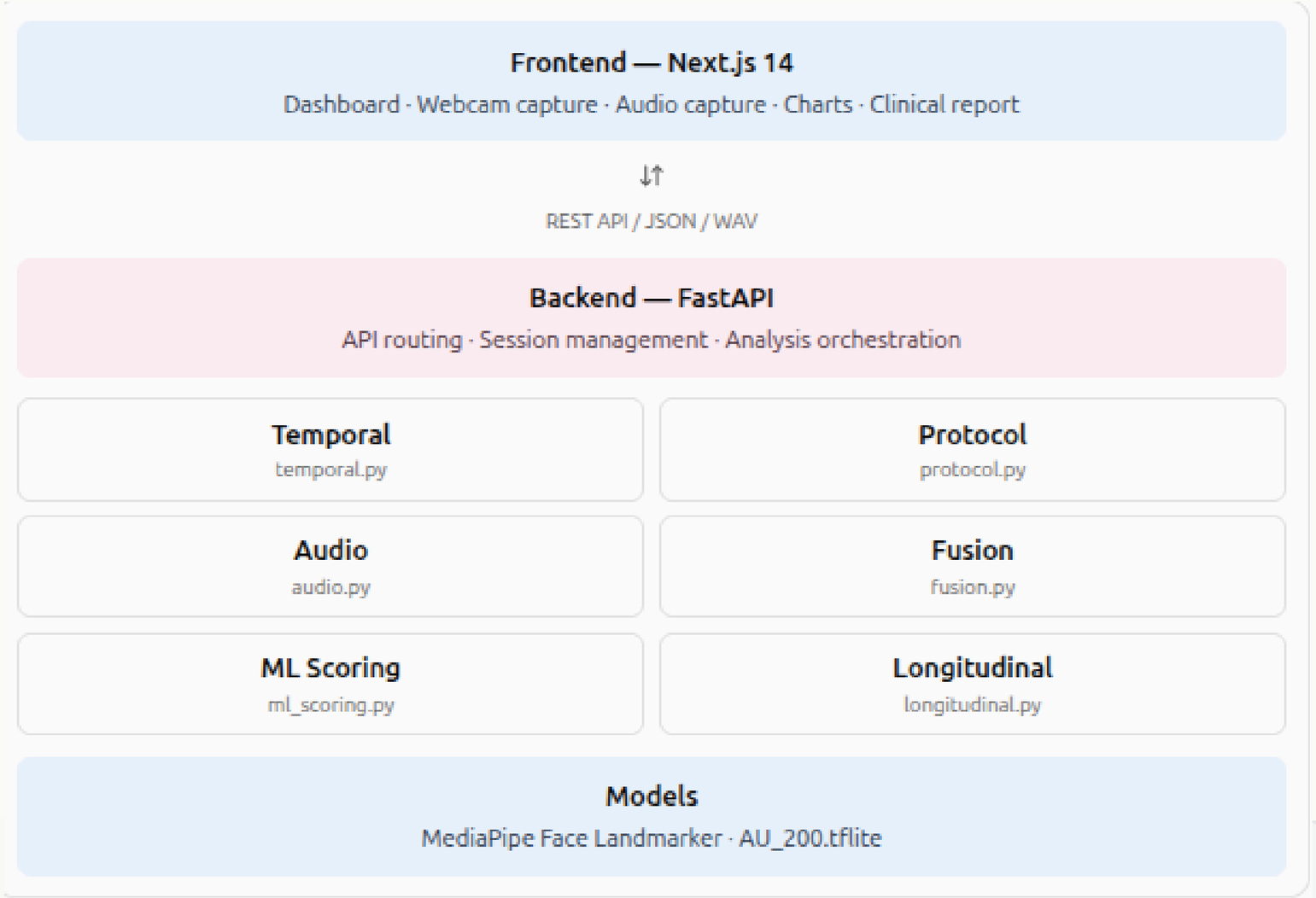
Depression interview corpus with multimodal behavioral signals and PHQ-8 labels.

AVEC

Affective computing benchmarks supporting comparable multimodal feature research.

Statistical functionals and synthetic calibration reference these datasets’ conventions.

System Architecture



Main Software Modules

main.py

API bootstrap + routing

temporal.py

smoothing + calibration

protocol.py

guided assessment state

audio.py

acoustic extraction

fusion.py

feature integration

ml_scoring.py

prediction + attribution

longitudinal.py

history + export

test_pipeline.py

synthetic evaluation

API Implementation

POST	`/analyze · /analyze-base64`	single-frame analysis
POST	`/temporal/frame · /calibrate`	temporal state
POST	`/protocol/start · /frame · /audio`	guided session
POST	`/protocol/report · /ml/predict`	report + scoring
GET	`/longitudinal/history · /export`	records + export
DELETE	`/longitudinal/purge`	privacy purge

FastAPI orchestrates capture, session state, analysis and privacy operations through a consistent REST surface.

FacialPulse

Clinical Affect & Stimulus-Response Depression Screening

HIPAA / GDPR Compliant Zero Raw Media

Longitudinal History 0

12:02:12
Saturday, November 14, 2026

Morphological Baseline: Active Resting Bias Normalized
All expressions normalized against your natural resting morphology

Voice Active

Reset Baseline

Start Guided 4-Phase Screening

SCREENING PROTOCOL

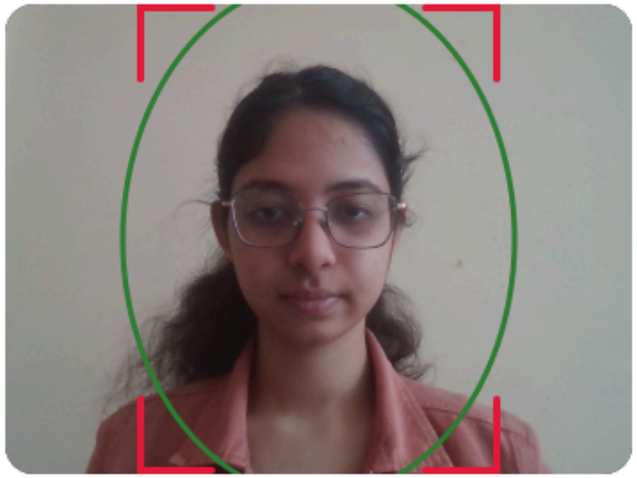
1. Resting Baseline
Captures resting facial morphology to eliminate false-positive bias

2. Positive Elicitation
Evaluates anhedonia & Duchenne smile latency

3. Cognitive Reading
Assesses downward gaze dwell & corrugator brow tension

4. Affective Recovery
Measures return rate to baseline & negative carryover

Face Detected & Tracking



Scan Now | Auto Monitor | ↺ Reset | Stop

Clinical Affect Analysis

Temporal Biomarker Risk
Window: 2 frames • Raw

25%
Low / Minimal Risk

Affect Flatness / Immobility Index: 0.66 / 1.00

Brow Tension Dwell 100%	Lip Depression Dwell 0%
Head Down Dwell 0%	Peak Smile Response 63%

Temporal Findings:

- Persistent brow tension (100% of window)

Expression Dynamics Trajectory — Last 2 updates

FacialPulse

Clinical Affect & Stimulus-Response Depression Screening

HIPAA / GDPR Compliant Zero Raw Media

Longitudinal History 0

12:05:52
Saturday, November 14, 2026

Morphological Baseline: Active Resting Bias Normalized
All expressions normalized against your natural resting morphology

Voice Active

Cancel Protocol

STEP 3 OF 4 **Phase 3: Standardized Reading Task**

9s remaining

Baseline (8s) | Positive Affect (12s) | Reading Task (15s) | Recovery (8s)

Read the following passage aloud or quietly to yourself:

> *"When sunlight strikes raindrops in the air, they act as a prism and form a rainbow. The rainbow is a division of white light into many beautiful colors."*

Voice Detected — Reading Passage in Progress
Capturing fundamental frequency (F0), pause-to-speech ratio, and vocal perturbation

SCREENING PROTOCOL

1. Resting Baseline

Phase 3: Standardized Reading Task (9s)

Action Units
Clinical Indicators

AU01 Inner Brow Raiser 0.2%

MILD AFFECTIVE BLUNTING ⚡ MULTIMODAL FUSION: ACTIVE Session ID: fp_1794638125

Guided 4-Phase Affective & Multimodal Screening Report

Multimodal assessment combined 4-phase facial analysis (Visual Risk: 14.0) with clinical acoustic speech prosody (Acoustic Risk: 56.7). Overall Cross-Modal Concordance is 57%. Symptom profiling indicates mild affective blunting with Flatness Index at 24.7/100 and Psychomotor Pause Retardation at 55.0/100.

30%

Composite Risk Score

Visual: 14% • Acoustic: 56.7%

🔗 **Cross-Modal Concordance: 57.3% (Moderate Confidence)**

Affective Incongruence: Masked Facial Presentation (Speech displays pronounced psychomotor blunting despite responsive facial dynamics)

AUDIO-VISUAL SYNCHRONIZED

Affective Flatness 25%	Psychomotor Retardation 55%	Anhedonia / Positive Deficit -31%	Cognitive Tension 40%
Blunted Facial Expressivity & Acoustic Monotone Prosody	Head-Down Dwell (0%) & Speech Pause Latency	Peak Smile: 83% • Duchenne: 0%	AU04 Brow Furrow (0%) & Vocal Strain

🎙️ Acoustic Speech Biomarkers (Rainbow Passage Analysis)

Duration: 44.97s • Sample Rate: 16 kHz

Pitch Variation (F0 Std)	Speech Pause Ratio	Avg Pause Duration	Vocal Strain / Jitter
78.7 Hz	**99%**	**4932 ms**	**100%**
Normal Inflection	Elevated Silence	Prolonged Pauses	Perturbation Alert

ML Depression Calibration & PHQ-8 Benchmark Assessment

Trained on DAIC-WOZ & AVEC affective computing benchmarks using Gradient-Boosted ensemble stacking

⚠️ DSM-5 CLINICAL CUTOFF MET (≥10)

10 / 24 — Moderate

Continuous: 9.91 • 95% CI: [7.7, 12.1]

Exceeds standard clinical threshold (>= 10); formal psychiatric clinical evaluation advised.

Minimal (0-4) | Mild (5-9) | Moderate (10-14) ★ Cutoff | Mod. Severe (15-19) | Severe (20-24)

0 — 5 — 10 (DSM-5 Cutoff) — 15 — 20 — 24

Item-Level DSM-5 Diagnostic Mapping (Observable Physical Criteria):

ITEM 1 — 0 / 3

Loss of Pleasure / Anhedonia

Not at all (0)

Smile Peak: 83%, Duchenne Co-activation: 0%

ITEM 2 — 0 / 3

Depressed / Downcast Mood

Not at all (0)

Residual Sadness: +0.00, Brow Lowerer: 0%

ITEM 4 — 2 / 3

Energy Deficit / Blunted Affect

More than half (+2)

Hypomimia Index: 21%, Acoustic Energy Flatness: 45%

ITEM 7 — 3 / 3

Cognitive Burden & Tension

Nearly every day (+3)

Corrugator AU04 Dwell: 0%, Vocal Jitter: 21.09%

ITEM 8 — 3 / 3

Psychomotor & Speech Retardation

Nearly every day (+3)

Head-Down Dwell: 0%, Pause Ratio: 99%

Key Biomarker Attributions (Top Contributors to Machine Learning Prediction):

avg pause duration ms: +730.67 | monotone index: -146.47 | vocal strain index: +5.01 | speech pause ratio: -4.68 | affect flatness index: +4.45

Longitudinal Trajectory & Session-over-Session Tracking

SUBJ-2CEABB5A6C — Baseline Session

Δ vs Previous Session

Δ vs Initial Baseline

Clinical Criteria (DSM-5 / APA)

Longitudinal Trajectory & Session-over-Session Tracking

SUBJ-2CEABB5A6C

Baseline Session

Δ vs Previous Session	Δ vs Initial Baseline	Clinical Criteria (DSM-5 / APA)
0 pts	**0 pts**	**±5 Points Cutoff**
Stable Trajectory	Cumulative shift across all sessions	Accepted threshold for meaningful response

Key Affective & Psychomotor Findings:

- Positive smile reactivity and baseline affective recovery were fully preserved.
- Psychomotor speech retardation: pause ratio 99% with avg pause duration 4932.2ms.
- Acoustic vocal perturbation/strain observed during reading passage (Index: 100.0).
- ML Benchmark Calibration (DAIC-WOZ): Predicted PHQ-8 is 10/24 (Moderate), meets DSM-5 clinical screening threshold (>= 10).

Multi-Phase Stimulus-Response Trajectory (Full 43s Session)



View Longitudinal History | Export De-Identified Record (JSON) | Print Clinical EHR Summary | Close & Return to Live Stream

References

1. Project repository — Suvanwita/IVP-FacialPulse
2. DAIC-WOZ depression interview corpus
3. AVEC affective computing benchmarks
4. MediaPipe Face Landmarker documentation
5. TensorFlow Lite documentation
6. FastAPI and Next.js 14 documentation
7. PHQ-8 scoring reference

THANK **YOU**

Thank you for your attention.